

Part I: Divide and Conquer

Lecture 5: Deterministic Linear-Time Selection

Computer Science and Technology
United International College

Objective and Outline

Objective:

- We discussed a **randomized** selection algorithm that runs in $\mathcal{O}(n)$ on **average**
- We now discuss a **deterministic** algorithm that runs in $\mathcal{O}(n)$ in the **worst** case

Reference: Section 9.3 of CLRS

Outline:

- 1 Idea and Algorithm
- 2 Analysis of Algorithm

Deterministic Selection Algorithm

Idea:

Question

If we follow the Partition idea to solve the selection problem, which step(s) make the worst case running time becomes $\mathcal{O}(n^2)$?

Ans: We make a “**bad**” split in each iteration

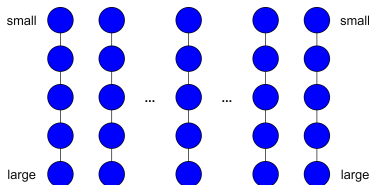
Trick: In each iteration, we pick a **good** element such that it “guarantees” a good split

Question

How to get such a “good” element in each iteration?

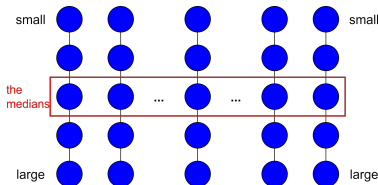
DSelection(A, p, r, i)

- 1 Divide the $n = r - p + 1$ items into $\lceil n/5 \rceil$ sets in which each, except possibly the last, contains 5 items. $\mathcal{O}(n)$



Algorithm illustration (step1): make groups of five

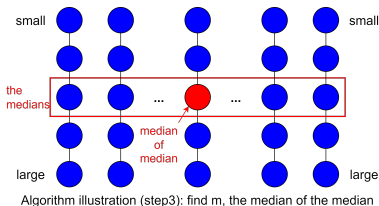
- 2 Find the median of each of the $\lceil n/5 \rceil$ sets. $\mathcal{O}(n)$



Algorithm illustration (step2): find the medians of $n/5$ group

DSelection(A, p, r, i)

- 3 Take these $\lceil n/5 \rceil$ medians and put them in another array. Use DSelection() to **recursively** calculate the **median** of these medians. Call this x . $T(n/5)$



Example

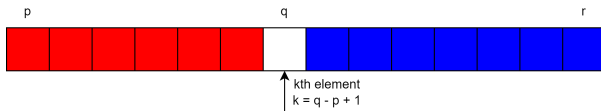
Find a good pivot for following list of numbers:

8 25 2 14 3 20 15 13 12 11 7 5 9 10 16 4 19 14 17 6
18 22 1 23 26 21

Do this on board.

DSelection(A, p, r, i)

4 Partition the original array using x as the pivot.



Let q be index of x , i.e., x is the $(k = q - p + 1)$ -th smallest element in original array. $\mathcal{O}(n)$

- 1 If $i = k$ return x
If $i < k$ return $DSelection(A, p, q - 1, i)$
If $i > k$ return $DSelection(A, q + 1, r, i - k)$
 $T(\max(q - p, r - q))$

Termination condition: If $n \leq 5$ sort the items and return the i th largest

The algorithm returns the correct answer, because

- Steps 4 and 5 will always return correct solution, **no matter** which x is used as pivot

The reason for Steps 1, 2, and 3 is to

- guarantee that x is “**near**” the center of the array $\Rightarrow$ a “good” split

1 Idea and Algorithm

2 Analysis of Algorithm

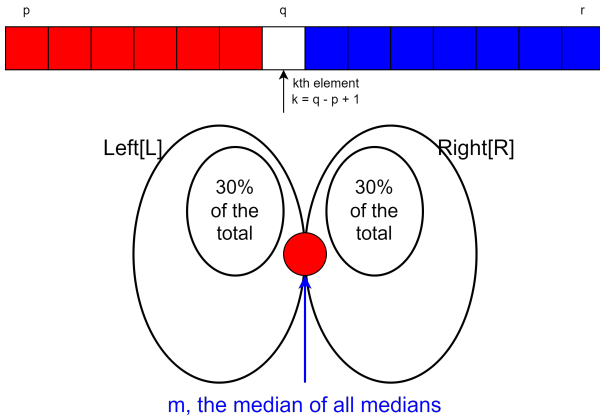
Running time:

$$T(n) \leq \mathcal{O}(n) + T(n/5) + T(\max\{k-1, n-k\})$$

Running Time

Lemma 5.1

At Step 4, at least $\frac{3n}{10} - 6$ elements are greater (less) than x .



Proof.

- Consider the $\lceil \frac{n}{5} \rceil$ groups and their medians.
- $\frac{1}{2}$ of those medians $\geq x$.
- Let A be the set of groups whose medians $\geq x$.
- Then, $|A| \geq \frac{1}{2} \lceil \frac{n}{5} \rceil$
- Let A' be obtained from A by removing the group with fewer than 5 items and the group that contains x .

$$|A'| \geq |A| - 2 \geq \frac{1}{2} \lceil \frac{n}{5} \rceil - 2$$

Proof.(Con't)

- Each group in A' :
 - ① Has 5 items.
 - ② Median $> x$.

Hence, each set has at least 3 items $> x$.

- Total number of items $> x$ is at least

$$3|A'| \geq 3\left(\frac{1}{2}\lceil \frac{n}{5} \rceil - 2\right) \geq \frac{3n}{10} - 6$$

Analysis of number less than x is exactly the same!

Running Time

- Assume that $T(n)$ is non-decreasing. Because of Lemma 5.1, the time used by step 5 is at most

$$T\left(n - \left(\frac{3n}{10} - 6\right)\right) = T\left(\frac{7n}{10} + 6\right)$$

- For technical convenience, we say that DSelection uses $\Theta(1)$ time for input with $n \leq 140$.
- Steps 1,2,4 take $\mathcal{O}(n)$ time. Suppose they take at most an time for some constant a .
- We have

$$T(n) \leq \begin{cases} \Theta(1) & \text{if } n \leq 140 \\ T(\lceil n/5 \rceil) + T(7n/10 + 6) + an & \text{if } n > 140 \end{cases}$$

We will show by strong induction on n that

Theorem

$T(n) \leq cn$ for some constant c and $\forall n$

Proof.

Induction basis: $\forall n \leq 140$ We can make c large enough that

$$T(n) \leq cn \quad \forall n \leq 140$$

Inductive hypothesis: Inequality is true for all $n < N$

Induction step: Now consider the case of $n = N$:

$$\begin{aligned} T(n) &\leq T(\lceil n/5 \rceil) + T(7n/10 + 6) + an \\ &\leq c(\lceil n/5 \rceil) + c(7n/10 + 6) + an \text{ (induction hypothesis)} \\ &\leq cn/5 + c + 7cn/10 + 6c + an \\ &= 9cn/10 + 7c + an = \textcolor{red}{cn} + (-cn/10 + 7c + an) \end{aligned}$$

Proof.(Con't)

We can make c large enough such that $c \geq 20a$. Then:

$$\begin{aligned} T(n) &\leq cn + (-cn/10 + 7c + an) \\ &= cn + \frac{-(n-70)}{n}cn/10 + an \\ &\leq cn + \frac{-1}{2}cn/10 + an \text{ (because } n \geq 140\text{)} \\ &\leq cn + \frac{-1}{2}20an/10 + an \text{ (because } c \geq 20a\text{)} \\ &= cn \end{aligned}$$

Inequality is hence proved. And we have

$$T(n) = \mathcal{O}(n)$$

Question

In DSelection, input elements are divided into groups of 5. Will the algorithm work also in linear time if they are divided in to groups of 3? How about 7? Why not 4 or 6?